

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

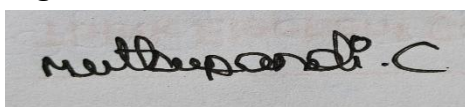
Concept Mapping
Annexure A

Criteria	Concept Identification (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _192565037 / Muthupandi C_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Annexure A

Concept Map 1: Decision Support Systems (DSS)

Task Description:

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decision-making.

What your map must show:

1. Core components: Data, Model, User Interface
 2. Types of DSS
 3. Flow: Data → Analysis → Decision
 4. Role in business decision-making
 5. Real-world example linkage
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Concept Map 2: Operational Systems vs DSS

Task Description:

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).

What your map must show:

1. Definitions of OLTP and DSS
2. Differences (data type, purpose, users, processing)
3. Similarities and interactions
4. Real-time vs historical data
5. Example systems (banking vs analytics)

Concept Map 3: Architecture of Data Warehouse

Task Description:

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.

What your map must show:

1. Data Sources → ETL → Data Warehouse → Data Marts
2. Metadata layer
3. Front-end tools (reporting, OLAP)
4. Data flow direction
5. Role of each layer

Concept Map 4: OLAP, Data Cubes & Dimensional Modeling

Task Description:

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modeling techniques.

🌀 What your map must show:

1. Data cube structure (dimensions, measures)
2. OLAP operations (slice, dice, roll-up, drill-down)
3. Star schema and snowflake schema
4. Fact and dimension tables
5. Analytical usage

Concept Map 5: Query Processing & BI Tools (KNIME)

Task Description:

Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.

What your map must show:

1. Query processing stages (Parsing, Optimization, Execution)
2. Role of BI tools
3. KNIME workflow structure
4. Data → Query → Visualization pipeline
5. OLAP application integratio

1. INTRODUCTION TO DATAWAREHOUSING

A Data Warehouse is a centralized repository that stores integrated, historical, and subject-oriented data from multiple sources to support business decision making.

Key Characteristics

- Subject-Oriented
- Integrated
- Time-Variant
- Non-Volatile



2. DECISION SUPPORT SYSTEMS (DSS)

DSS helps organizations make informed decisions using reports, analytics, dashboards, and data exploration.



3. OPERATIONAL vs DECISION SUPPORT SYSTEM

Operational System (OLTP)

- Day-to-day operations
- Handles current data
- Frequent updates
- Detailed data
- High volume transactions
- Examples: Banking, Booking Systems

Decision Support System (OLAP)

- Decision making & analysis
- Historical & summarized data
- Infrequent updates
- Aggregated data
- Complex queries
- Examples: Sales Analysis, Trend Analysis

VS

4. ARCHITECTURE OF DATAWAREHOUSE

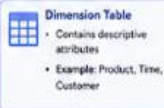


Components

- Data Sources**: Operational DB, Flat files, APIs, Logs, etc.
- ETL Process**: Extract, Transform, Load data
- Data Warehouse**: Central repository
- OLAP Server**: Multidimensional view & analysis
- Front End**: Reports, Dashboards, Visualization Tools

7. DIMENSIONAL DATA MODELING

Dimensional modeling is used to design the data warehouse for faster query and easier understanding.



Benefits

- Simpler structure
- Improved performance
- User friendly

INTRODUCTION TO DATAWAREHOUSING

Turning Data into Decisions

5. ETL (EXTRACT, TRANSFORM, LOAD)



ETL Benefits

- Improves data quality
- Consolidates data
- Ensures consistency
- Enables historical data storage

8. STAR AND SNOWFLAKE SCHEMAS



- Denormalized structure
- Faster query performance
- Simpler design



- Normalized structure
- Saves storage space
- More complex queries

6. OLAP AND DATA CUBES

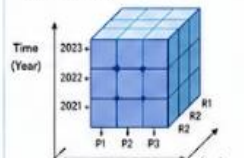
OLAP (Online Analytical Processing)

Allows users to analyze data from multiple dimensions.

- Fast query response
- Complex calculations
- Support for decision making

DATA CUBE

Multidimensional structure that organizes data in the form of a cube.



9. QUERY PROCESSING



10. OPEN SOURCE BUSINESS INTELLIGENCE TOOLS



HOW DATAWAREHOUSE WORKS ?



KEY BENEFITS OF DATAWAREHOUSING



APPLICATIONS

- Sales & Marketing Analysis
- Financial Reporting
- Customer Behavior Analysis
- Supply Chain Analysis
- Risk Management